

From Particles to Waves

Towards Coupling Dynamic
Snow Modelling and Spectral
Elements to Detect
Avalanche Breakoff in DAS
Data

Salome Bachmann

August 4, 2026



Overview

- Created 3 different models and 2 scenarios for crack propagation in snow using SALVUS
- Two pure spectral element (SE) models, one combined with data from material point method (MPM) simulations
- **Goal:** Couple dynamic snow modelling with SE to model crack propagation in snow

Why model crack propagation in snow? - Relevance

- 22 fatalities by avalanche in Switzerland every year (SLF, 2025)
- Avalanches are danger for human life but also infrastructure (Schweizer et al., 2003)
- Understanding processes associated with avalanches important
- Avalanche release because of loading, leading to crack propagation in a weak layer (Gaume et al., 2017)

Relevance

- Crack propagation at different speeds: sub-Rayleigh (below vs of material) and Supershear (above vs) (Trottet et al., 2022), transition between two regimes possible (Bergfeld et al., 2025)
- Fast crack propagation → critical crack length for release (Gaume et al., 2015) is reached faster
- Distributed Acoustic Sensing (DAS) already used to monitor avalanches in Switzerland (Edme et al., 2023; Paitz et al., 2023) and Norway (Turquet et al., 2024)
- No knowledge if crack propagation associated with breakoff can be seen in DAS data

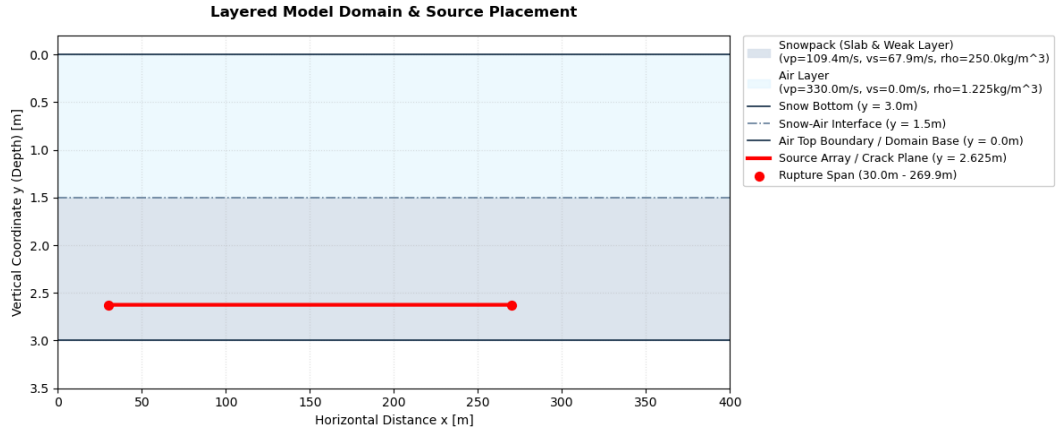
Research Questions

- What FE framework is required to to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?
- To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?
- How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?

Research Questions

- What FE framework is required to to simulate DAS measurements of crack propagation in snow, and what are the key assumptions and limitations?
- **What physical phenomena are observed in a two-layer model with a simple STF and how do they differ for sub-Rayleigh versus Supershear crack propagation?**
- **To what extent do the results change if source wavelets from a MPM simulation of a propagation saw test (PST) are injected?**
- **How could the phenomena observed be used to distinguish crack propagation regimes in real-world DAS data and aid in detecting snow slab avalanche breakoff?**

Simple Model I



Simple Model II

Figure: Two-layer model setup in SALVUS. Colors show materials, material types and properties in legend. Source line extent in red.

Bibliography I



Bergfeld, B., Gaume, J., Bobillier, G., Pellet, A., Schweizer, J., & van Herwijnen, A. (2025). Signatures of the sub-Rayleigh to supershear fracture transition in snow avalanche experiments. *Nature Communications*, 16(1), 11153.
<https://doi.org/10.1038/s41467-025-65825-6>



Edme, P., Paitz, P., Walter, F., van Herwijnen, A., & Fichtner, A. (2023, February). Fiber-optic detection of snow avalanches using telecommunication infrastructure.
<https://doi.org/10.48550/arXiv.2302.12649>



Gaume, J., van Herwijnen, A., Chambon, G., Birkeland, K. W., & Schweizer, J. (2015). Modeling of crack propagation in weak snowpack layers using the discrete element method. *The Cryosphere*, 9(5), 1915–1932.
<https://doi.org/10.5194/tc-9-1915-2015>

Bibliography II



Gaume, J., van Herwijnen, A., Chambon, G., Wever, N., & Schweizer, J. (2017). Snow fracture in relation to slab avalanche release: Critical state for the onset of crack propagation. *The Cryosphere*, 11(1), 217–228.
<https://doi.org/10.5194/tc-11-217-2017>



Paitz, P., Lindner, N., Edme, P., Huguenin, P., Hohl, M., Sovilla, B., Walter, F., & Fichtner, A. (2023). Phenomenology of Avalanche Recordings From Distributed Acoustic Sensing. *Journal of Geophysical Research: Earth Surface*, 128(5), e2022JF007011. <https://doi.org/10.1029/2022JF007011>



Schweizer, J., Bruce Jamieson, J., & Schneebeli, M. (2003). Snow avalanche formation. *Reviews of Geophysics*, 41(4). <https://doi.org/10.1029/2002RG000123>



SLF. (2025). Avalanche Victims since 1936 - Long Term Statistics.

Bibliography III



Trottet, B., Simenhois, R., Bobillier, G., Bergfeld, B., van Herwijnen, A., Jiang, C., & Gaume, J. (2022). Transition from sub-Rayleigh anticrack to supershear crack propagation in snow avalanches. *Nature Physics*, 18(9), 1094–1098.
<https://doi.org/10.1038/s41567-022-01662-4>



Turquet, A., Wuestefeld, A., Svendsen, G. K., Nyhammer, F. K., Nilsen, E. L., Persson, A. P.-O., & Refsum, V. (2024). Automated Snow Avalanche Monitoring and Alert System Using Distributed Acoustic Sensing in Norway. *GeoHazards*, 5(4), 1326–1345. <https://doi.org/10.3390/geohazards5040063>



Salome Bachmann
Master Student
sbachmann@student.ethz.ch

ETH Zurich
D-EAPS